The Influence of Dump Gap on External Combustor Aerodynamics at High Fuel Injector Flow Rates

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The increasing demand to reduce fuel burn, hence CO_2 emissions, from the gas turbine requires efficient diffusion to reduce the system pressure loss in the combustor. However, interactions between prediffuser and combustor can have a significant effect on diffuser performance. For example, the consequence of increased fuel injector flow at a dump gap set using conventional design guidelines has been shown (Walker, A. D., Carrotte, J. F., and McGuirk, J. J., 2007. "Compressor/Diffuser/Combustor Aerodynamic Interactions in Lean Module Combustors," ASME Turbo Expo 2007—Power for Land Sea and Air, Paper No. GT2007-27872) to introduce a destabilizing interaction between fuel injector and upstream components. The present paper concentrates on examining the effects of increased dump gap. Dump gap ratios of 0.8, 1.2, and 1.6 were employed, with each test utilizing the same inlet guide vane, compressor rotor, integrated outlet guide vane (OGV)/prediffuser, and dump geometry. The flow fraction of compressor efflux entering the combustor cowl was set to be representative of lean combustors (50–70%). Measurements were made on a fully annular rig using a generic flame tube with metered cowl and inner/outer annulus flows. The results demonstrate that, with fixed cowl flow, as the dump gap increases, component interactions decrease. At a dump gap ratio of 0.8, the proximity of the flame tube influences the prediffuser providing a beneficial blockage effect. However, if increased to 1.2, this beneficial effect is weakened and the prediffuser flow deteriorates. With further increase to 1.6, the prediffuser shows strong evidence of separation. Hence, at the dump gaps probably required for lean module injectors, it is unlikely the prediffuser will be influenced beneficially by the flame tube blockage; this must be taken into account in the design. Furthermore, with small dump gaps and high cowl flow fraction, the circumferential variation in cowl flow can feed upstream and cause OGV/rotor forcing. At larger dump gaps, the circumferential variation does not penetrate upstream to the OGV, and the rotor is unaffected. The optimum dump gap and prediffuser design for best overall aerodynamic system performance from rotor through to feed annuli is a compromise between taking maximum advantage of upstream blockage effects and minimizing any 3D upstream forcing. [DOI: 10.1115/1.3028230]

Introduction

In future aeroengine gas turbine combustor designs, lean module combustion will be essential in order to achieve required NO_x targets. Consequently, up to 70% of the flow issuing from the compressor may be required to enter the combustor cowl, compared with (typically) 30% in current combustor designs. To accommodate this increased mass flow requirement while maintaining an acceptable pressure drop, lean module fuel injectors will be significantly larger than their conventional air-spray counterparts. This will necessitate geometric changes to the combustor external aerodynamic layout, such as a deeper flame tube and larger dump gaps, potentially increasing the aerodynamic loss due to increased turning within the dump region. Additionally, these geometry and flow split changes and the strongly 3D (circumferential) distribution of the flow into the combustor head compared to current practice are likely to increase the coupling or interaction between the combustor external aerodynamics and upstream components such as the compressor outlet guide vane (OGV)/prediffuser combination and possibly even lead to forcing of the final compressor rotor stage. Indeed previous work by Walker et al. [1] showed that it may be necessary to adopt an integrated design approach in order reduce the aerodynamic loss. By integrating the design of the OGV into the prediffuser and by introducing lean and sweep, the secondary flow structure at exit from the OGV was manipulated promoting radial transfer of higher momentum fluid into the diffuser end wall boundary layers. Not only did this enable the prediffuser area ratio to be increased to 1.8 from a conventional nonintegrated design of 1.6 but it also improved the prediffuser exit profile. The net result was an improvement in the overall flow uniformity and an estimated reduction in the system total pressure loss of approximately 20%. However, although this work was conducted using a generic lean module combustor geometry, the flow split was still typical of conventional systems. In a further piece of work, Walker et al. [2] investigated the effect of increasing the cowl flow from typical air-spray levels to those required for lean combustion (30% and 50–70% of compressor efflux, respectively). The results demonstrated that, with fixed geometry, as the injector flow increases, the performance of the prediffuser and feed annuli suffers. Prediffuser losses increased and at higher cowl flow rates the diffuser moved close to separation. Furthermore, the substantial circumferential variation in cowl flow fed upstream causing notable differences in performance to be observed inline and between injectors at OGV exit, ultimately causing rotor forcing.

The present investigation extends the studies of Walker et al. [1,2] to consideration of the dump gap. The earlier studies em-

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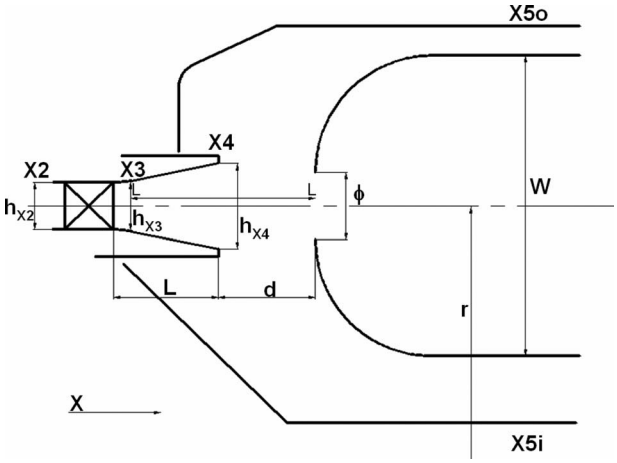

Fig. 1 Geometrical notation

ployed a fixed dump gap typical of conventional combustors. However, it is unlikely that dump gaps defined using existing design rules will be sufficient to accommodate the increased size of lean module fuel injectors. The dump gap is defined as the axial distance (d) from the prediffuser exit plane to the flame tube and is usually expressed as a fraction of the diffuser exit height, d/h_{X4} (for a definition of geometric parameters, see Fig. 1). As this ratio reduces, the proximity of the flame tube induces flow curvature within the prediffuser and a migration of flow toward the end walls. Hestermann [3] showed that this results in a more uniform diffuser exit profile and can suppress flow separation due to the introduction of higher momentum fluid into the boundary layers. Fishenden and Stevens [4] demonstrated that optimizing the dump gap also increases prediffuser pressure recovery but stated that it does not necessarily follow that this will result in lower losses to the feed annuli. Initially, as the dump gap reduces, the beneficial blockage effect increases diffuser performance and reduces the annulus losses. However, as the dump gap reduces beyond a certain point, the annulus loss begins to increase rapidly as the diffuser efflux must now turn in a much shorter space hence incurring a higher loss. Hestermann [3], Fishenden and Stevens [4], and Carrotte and Barker [5] all attempted to optimize the dump gap, in terms of diffuser and annulus performance, but for flame tube depths appropriate to traditional and not lean module combustors. Expressing the flame tube depth as a ratio of diffuser inlet height (W/h_{X3}), their results are summarized in Table 1 and it is noticeable that the optimum dump gap increases with flame tube depth. It is difficult to say how applicable these data are to lean module combustors, hence the current study was undertaken to characterize the extent that the inevitable change in dump gap will alter the external combustor aerodynamics. The experimental study utilizes the same rig as used by Walker et al. [1,2], which comprises a fully annular generic combustor downstream of a single stage axial compressor. The OGV/prediffuser coupling is effected via an advanced integrated OGV/prediffuser geometry [1]. Several traverse planes are used to gather five-hole probe data, which allow the flow structure to be examined through the rotor, OGV, prediffuser, and in the inner/outer annulus supply ducts, quantifying the extent of the component interactions and associated system performance. Several configurations have been investigated where the cowl mass flow has been set at typical lean combustion levels (50–70% of compressor efflux) and the dump gap ratio (d/h_{X4}) varied from 0.8 to 1.6.

Table 1 Optimum dump gap [3–5]

Flame tube depth (W/h_{X3})	Dump gap (d/h_{X4})
3.5	0.6
4.8	0.7–0.8
5.5	0.8–1.0
Lean module combustor >6.0	>1.0?

Experimental Facility

During this project, all experimental data were obtained on a low speed isothermal test facility operating at nominally atmospheric conditions. Figure 2 provides an illustrative sketch, and full details are provided in Ref. [1]. The working section (Fig. 3) comprises a single stage axial compressor (IGV and rotor), an OGV/prediffuser assembly, and a generic representation of a downstream combustion system. Geometric details of the IGV, rotor, and OGV rows are also given in Ref. [1]. The rotor is operated at a fixed nondimensional speed ($N\pi D/\sqrt{(\gamma RT)}$) and mass flow ($\dot{m}\sqrt{(RT)}/AP$), the mass flow being derived from measurements in the calibrated inlet section located upstream of the compressor. The operating conditions corresponded to a fixed flow coefficient $\phi(=V_a/U)$ of 0.403 and a mass flow of approximately 4 kg s^{-1} resulting in an axial velocity through the blade rows of approximately 40 m s^{-1} (Mach number 0.12) and a Reynolds number (based on OGV chord) of approximately 1.6×10^5. The downstream flow distribution to the feed annuli and combustor cowl is controlled by a system of throttles, and a centrifugal fan is utilized as a further throttle to ensure that the axial compressor maintains the desired operating condition. It should be noted that for all the geometry and flow split conditions studied experimentally here, the overall nondimensional rotor speed and mass flow operating conditions remained unchanged.

Test Geometry

Integrated OGV/Prediffuser. The OGV row has 160 vanes, whose geometry includes specific lean, sweep, and camber characteristics that were determined in Ref. [6] by a process of manual computational fluid dynamics (CFD) optimization. This achieves a spanwise blade loading variation that creates secondary flows, which transport higher momentum fluid toward the walls, thus offloading the end wall boundary layers within the prediffuser. Hence, as reported in Ref. [1], this integrated OGV (IOGV) design was able to achieve a prediffuser area ratio of 1.8 without separation while maintaining the same nondimensional length ($L/h_{\mathrm{in}}=2.23$). According to the standard diffuser loading chart of Howard et al. [7], a conventionally designed diffuser would have resulted in separation.

Generic Flame Tube, Cowl, and "Fuel Injectors". The same simple flame tube geometry as used previously by Walker et al. [1,2] was retained in the present work. This presents the IOGV/prediffuser with a representative downstream blockage. Lean module combustors may not feature air admission ports in the annuli, hence no holes were included in the flame tube liner walls. The resultant generic flame tube, illustrated in Fig. 4, has 20 simulated "fuel injectors" represented for convenience by 20 simple holes. The initial area of these holes was set in Ref. 2 to match the effective area of a traditional air-spray injector. As the mass flow through the cowl was increased, the ratio of cowl mass flow to hole area was maintained by increasing the size of the holes. This means that as the mass flow increased the relative acceleration of the dump flow into the cowl did not change. In the current study, cowl flows relevant to lean module levels were employed (50–70% of compressor efflux) with the remaining flow split equally between the annuli. Three dump gaps ratios (d/h_{X4}) were employed beginning with a value of 0.8 as employed in Refs. [1,2]. However, it is unlikely that such a dump gap would provide enough space for a lean module injector. Thus, the dump gap ratio was increased to 1.2 and 1.6 representing values that are large enough to accommodate a typical lean module injector.

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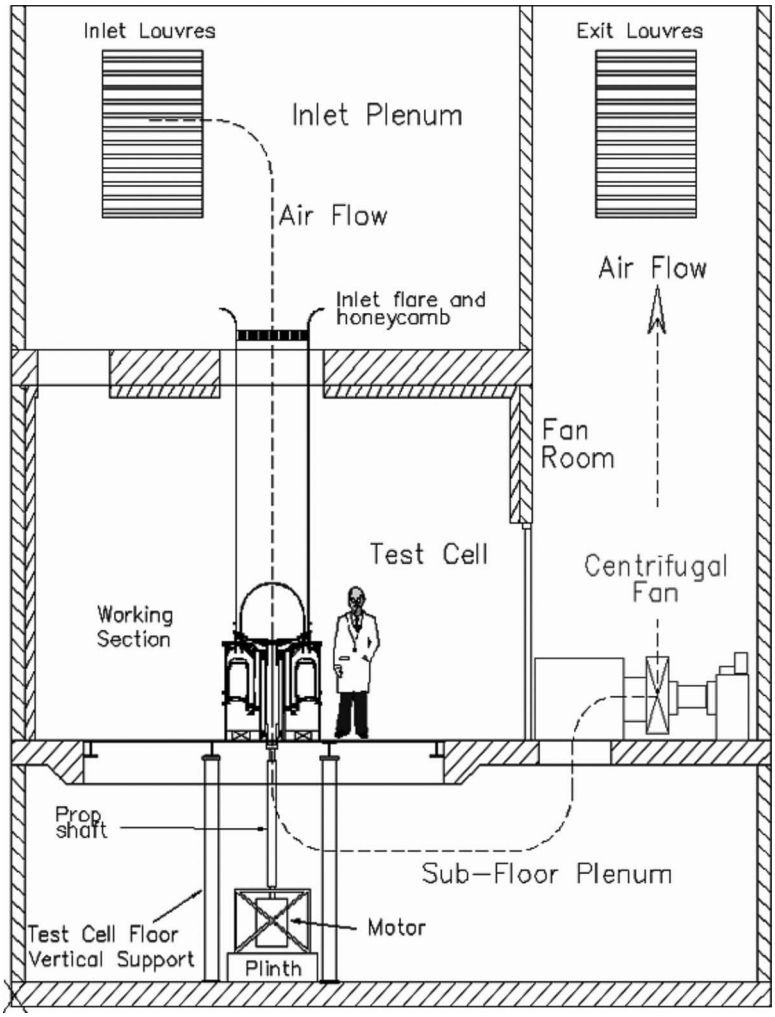

Fig. 2 Test facility

Instrumentation

Probes. Aerodynamic performance was assessed using suitably calibrated miniature five-hole probes as described by Wray and Carrotte [8] and employed in a non-nulled mode to determine the local flow vector and total and static pressures. As indicated in Fig. 3, five-hole probe area traverses could be conducted at rig inlet ($X1$), rotor exit ($X2$), OGV exit ($X3$), prediffuser exit ($X4$), within the inner ($X5_i$) and outer ($X5_o$) feed annuli and at various axial locations within the dump. The probes were traversed in the radial direction using a stepper motor powered linear guide attached to the external casings of the test section. Circumferential traversing was achieved by rotation of the casing or rotation of the IGV, OGV, and flame tube. The flow passage area traversed typically consisted of two OGV passages at stations close to the compressor (e.g., $X2$ and $X3$) but as whole combustor sector information was required further downstream (e.g., at prediffuser exit and within each feed annuli), the traverse area was typically increased to eight OGV passages (one combustor sector).

Data Reduction and Errors. Overall aerodynamic performance is quantified in terms of total pressure loss and static pressure recovery coefficients derived from the five-hole probe area traverses data, corrected to ICAO standard day conditions. Thus, at any measurement plane the mass flow rate and bulk average velocity are defined via

$$\dot{m} = \int \rho U dA = \rho \bar{U} A \tag{1}$$

The mass flow balance on area traverse planes was within $\pm 1\%$ of the rig mass flow entering the test facility measured via a five-hole probe traverse at plane ($X1$). Spatially averaged values of total and static pressures at each plane were derived using the mass-weighted technique described by Klein [9], e.g.,

$$\tilde{P} = \frac{1}{\dot{m}} \int P d\dot{m} = \frac{1}{\rho \bar{U} A} \int P \rho U dA \tag{2}$$

The mass-weighted total pressure loss (λ) and static pressure rise (Cp) coefficients are then defined as

$$\lambda_{1-2} = \frac{\tilde{P}_1 - \tilde{P}_2}{\tilde{P}_1 - p_1}, \quad Cp_{1-2} = \frac{\tilde{p}_2 - \tilde{p}_1}{\tilde{P}_1 - \tilde{p}_1} \tag{3}$$

The mass-weighted total pressures were repeatable to within ± 1 mm H_2O, which amounts to less than $\pm 0.5\%$ of the dynamic pressure at OGV exit. Thus, the repeatability of the derived total pressure loss and static pressure rise coefficients was better than ± 0.005 of the values presented.

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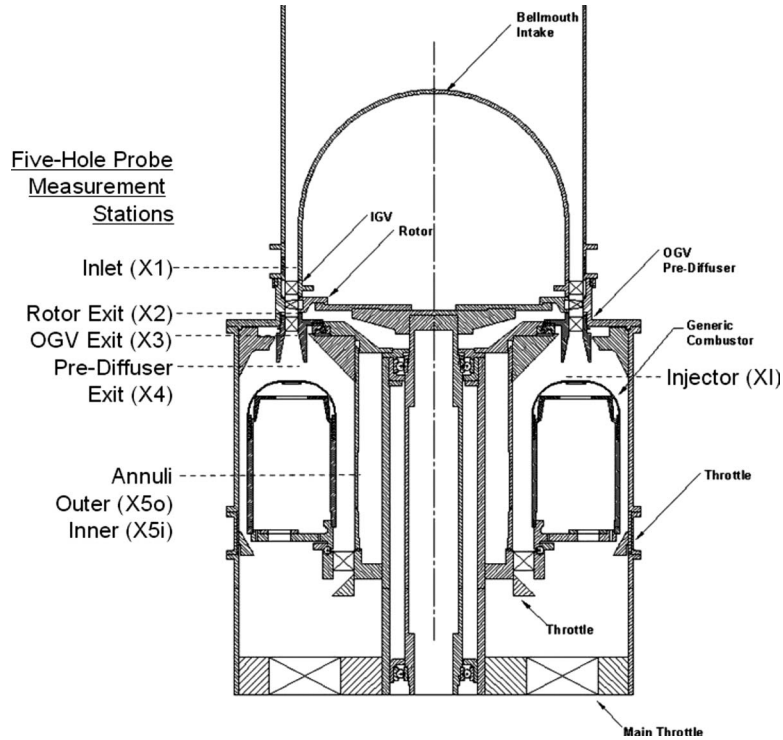

Fig. 3 Measurement planes

Results and Discussion

Prediffuser Exit ($X4$) and Dump Flow. Figure 5 presents normalized axial velocity contours over a full injector sector for dump gap ratios of 0.8, 1.2, and 1.6 and with the flow split entering the cowl set at 70% of the compressor efflux. OGV wakes are still clearly visible at prediffuser exit and exhibit only subtle differences as the dump gap ratio increases. However, the most notable change is the deterioration of the end wall boundary layer flow, particularly on the inner wall. The OGV/prediffuser was optimized for a conventional combustor, with a 30% cowl flow, in the previous study by Barker et al. [6]. Certainly it is shown here not to be the optimum design for the current configuration. With an increase in the cowl flow to 70%, while maintaining the dump gap (Fig. 5(a)), the inner wall shows some low velocity regions. But, as the flame tube moves away from diffuser exit, its influence reduces and the low velocity regions become larger and more distinct. Indeed, at the highest dump gap (Fig. 5(c)), the inner wall flow shows strong evidence of separation. The changing character of the flame tube blockage effect (acting to stabilize the prediffuser flow) as the cowl flow fraction increases is clearly illustrated in Fig. 6. This displays the prediffuser exit static pressure profile for a fixed dump gap ($d/h_{X4}=0.8$) for varying cowl flow (30%, 50%, and 70%). At the smallest cowl flow, the pressure profile has the expected "n" shape indicating a positive blockage effect—increasing the pressure at diffuser center relative to the walls, hence driving flow toward the end wall boundary layers. As the cowl flow increases, the suction effect eventually turns the pressure profile into a "u" shape (low pressure at diffuser center relative to the walls), which is not beneficial for end wall boundary layers to remain attached. Pitch-averaged profiles generated from sector data at the highest cowl flow (Fig. 7) further highlight the deterioration of the prediffuser flow with the end wall velocity dropping to less than 10% and 40% of the bulk mean for the inner and outer walls, respectively, at the larger dump gap. Given the data are time averaged it is thought that to avoid any transient separation within the prediffuser a value at the location nearest to the wall of above 40% is desirable. The pitch-averaged static pressure profiles demonstrate that, at this cowl flow, any beneficial back pressure generated by the flame tube is so weak relative to the suction effect that the u shaped pressure profile is present for all dump gaps. Static pressure contours (Fig. 8) also show this trend but reveal, at the smallest dump gap ratio of 0.8, a high

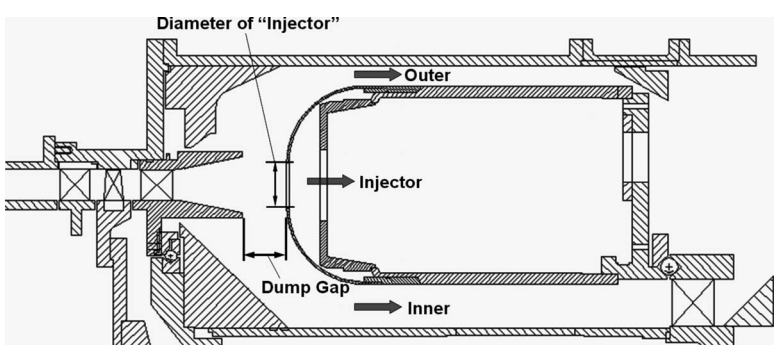

Fig. 4 Combustor details

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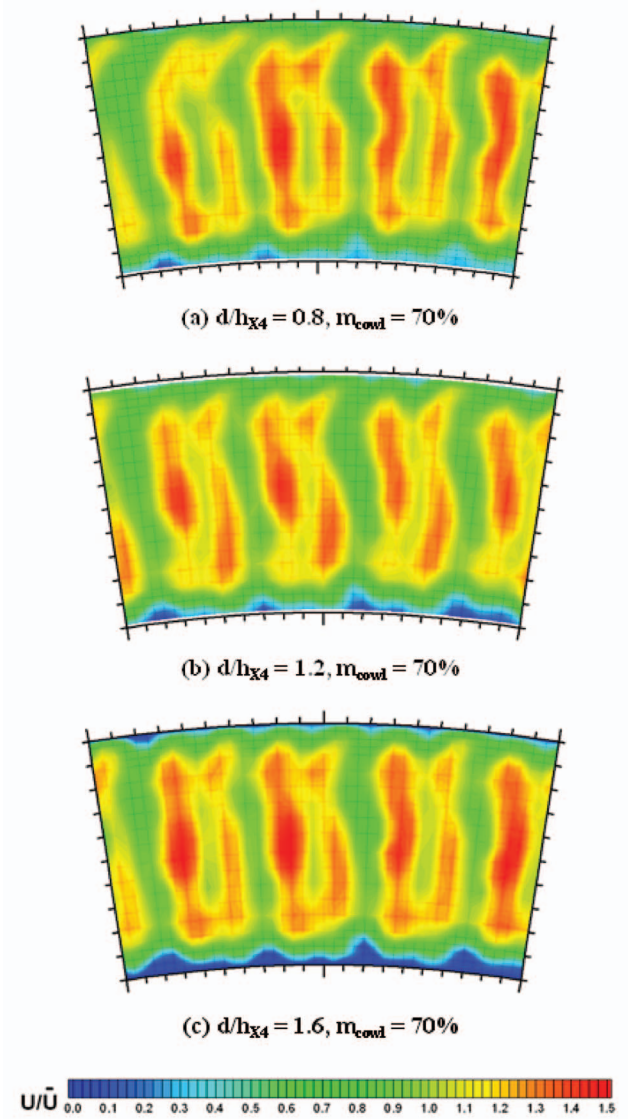

Fig. 5 Axial velocity contours at prediffuser exit (*X*4) (height 65.9 mm, sector angle $\Delta\theta$=18 deg, 8-OGV passages)

azimuthal variation caused by the circumferentially varying downstream pressure field on the combustor head from a high impingement value between cowl holes to a low suction value in line with the holes caused by the high (70%) cowl flow. Velocity data show that, with respect to the overall rig mass flow rate, the indicated mass flow per sector varies from inline to between injectors by $\pm 10\%$. However, as the dump gap ratio increases beyond 1.2, the circumferential static pressure variation decreases suggesting the prediffuser no longer feels any influence of the flame tube. In order to examine this phenomenon, further axial traverses were performed along the prediffuser centerline (line L-L in Fig. 4) inside the prediffuser and dump region at 50% and 70% cowl flows. Figure 9 plots the normalized static pressure for the three dump gap ratios both inline and between the injectors. Also shown on each plot is a numerically generated prediction for the prediffuser with no flame tube which provides data in which the prediffuser flow develops with no influence from flame tube blockage or circumferential variation due to cowl flow. For a dump gap ratio of 0.8, the static pressure inline and between injectors begins to diverge from the data at $d/h_2 \approx -0.35$, suggesting the flow within the prediffuser is influenced by the flame tube. As the dump gap ratio increases to 1.2 and 1.6, the static pressure

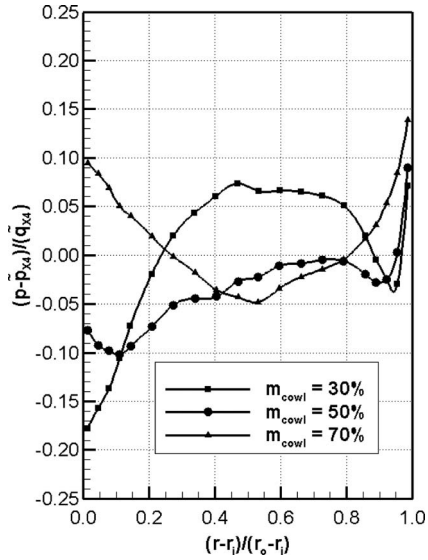

Fig. 6 Pitch average static pressure at prediffuser exit (*X*4), fixed dump (d/h_{X4}), varying flow

only begins to deviate downstream of diffuser exit at $d/h_2 \sim 0.1$ and 0.25, respectively, suggesting that the prediffuser, at least on its centerline, does not feel the any influence from the flame tube blockage. Figure 9 also serves to illustrate the circumferential variation caused by the flow through the injectors and the fact that

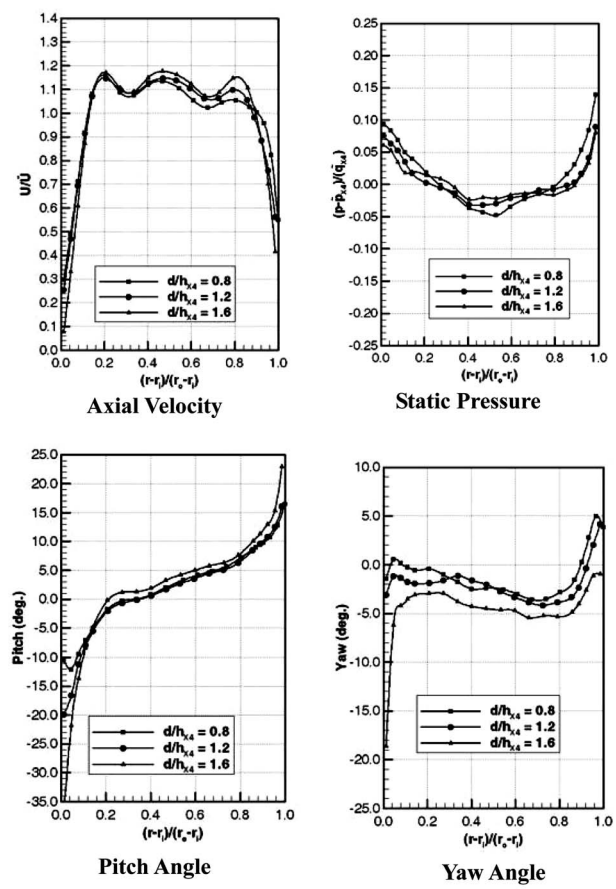

Fig. 7 Pitch averaged profiles at prediffuser exit (*X*4) injector mass flow, m_{cowl}=70%

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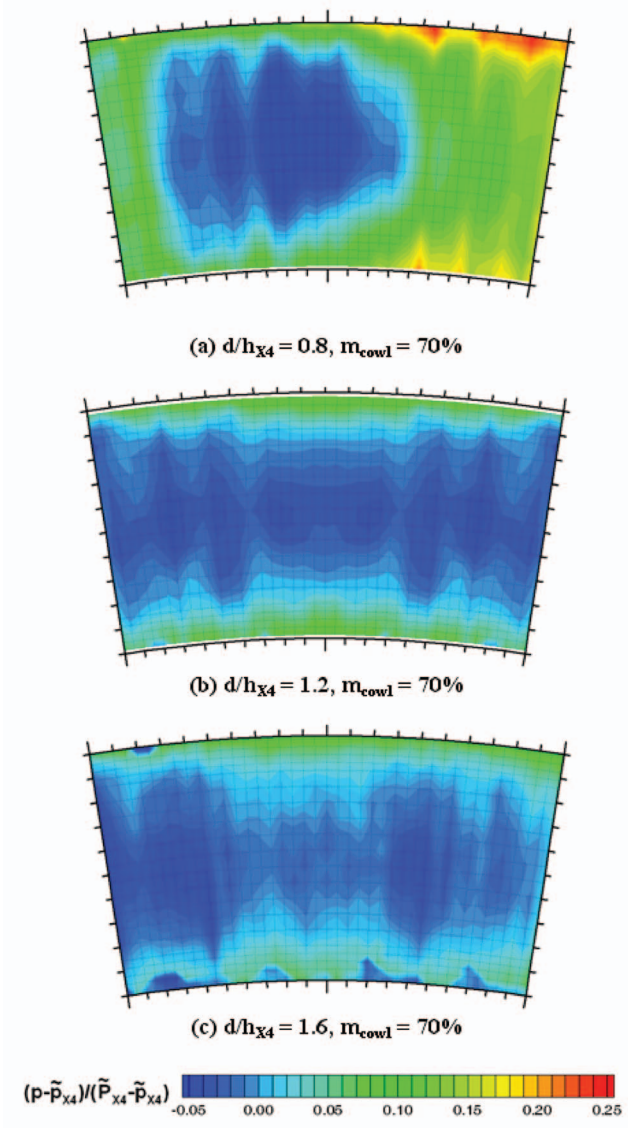

Fig. 8 Static pressure contorus at prediffuser exit (*X*4), (height 65.9 mm, sector angle $\Delta\theta$=18 deg, 8-OGV passages)

this penetrates further upstream at reduced dump gaps with a larger circumferential difference inline and between at the higher flow rates.

Mass-weighted static pressure recovery and total pressure loss coefficients (Table 2) for a complete injector sector with respect to rotor exit do not show a noticeable difference in performance for the two smaller dump gaps due to the fact that the low mass-flow near wall regions do not contribute significantly in the mass-weighted averaging process. However, at the larger dump gap, the near wall flow deterioration is indicated by a clear increase in λ and a decrease in Cp.

OGV Exit (*X*3). Axial velocity contours at a cowl mass flow of 70% are plotted for each dump gap in Fig. 10. Data were taken over a two OGV sector both inline and between injectors. Adjacent OGV wakes differ because there is one IGV per two OGV and the IGV wakes persists through the rotor, deepening one of the OGV wakes. Note that the OGV downstream of an IGV wake is the right-hand vane for the inline sector, but the left-hand vane for the between sector. For the two larger dump gaps, where the results above showed that the presence of the flame tube is not felt at the prediffuser exit, there is nominally no difference between

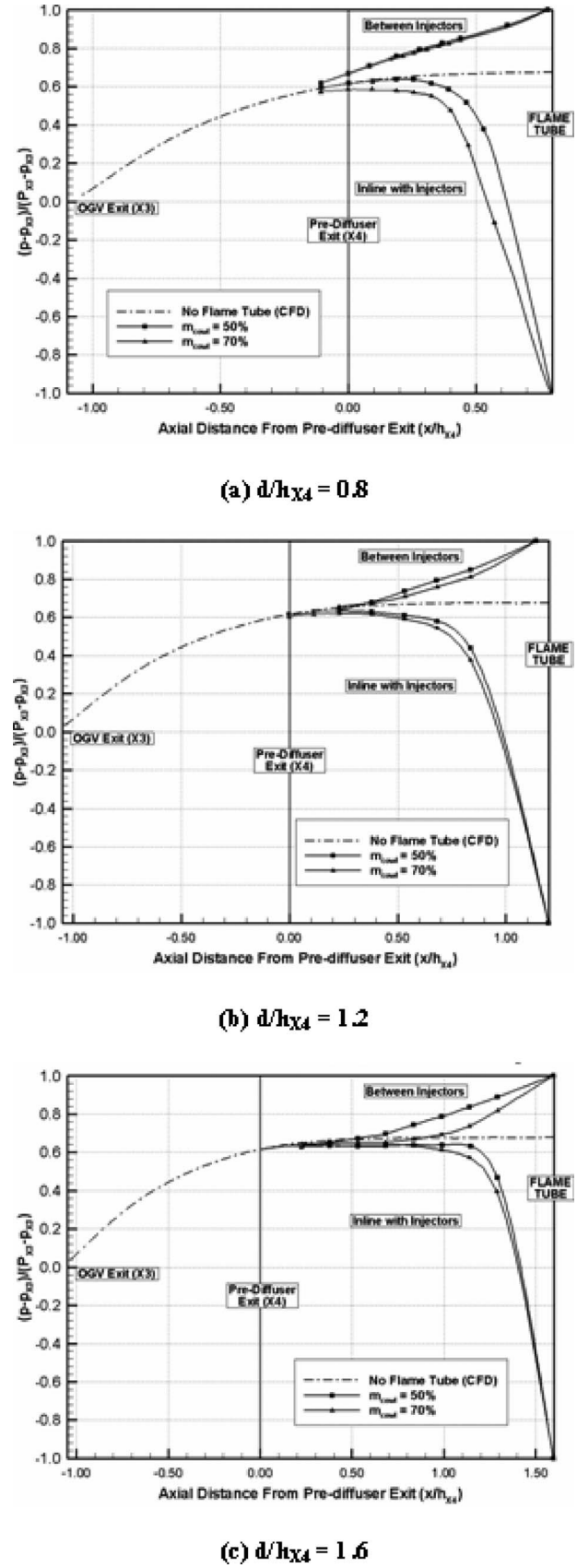

Fig. 9 Dump static pressure (on line L-L in Fig. 3)

the flow field inline and between the injectors. However, Fig. 10 shows that with the smallest dump gap there is a notable differ-

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Table 2 Prediffuser performance data (m_{cowl}=70%)

Dump gap (d/h_{X4})	λ_{X2-X4}	Cp_{X2-X4}
0.8	0.163	0.660
1.2	0.165	0.660
1.6	0.179	0.648

Table 3 OGV performance data (m_{cowl}=70%)

Dump gap (d/h_{X4})	λ_{X2-X3}	Cp_{X2-X3}
0.8 (2-OGV sector inline)	0.097	0.39
0.8 (2-OGV sector between)	0.153	0.36
0.8 (injector sector)	0.129	0.38
1.2	0.127	0.38
1.6	0.126	0.38

ence. Indeed the velocity data show that with respect to the overall rig mass flow rate, the indicated mass flow per sector varies form inline to between injectors by 10%. Table 3 details the total pressure loss and static pressure recovery across the OGV.

For the two larger dump gap ratios, performance is maintained. Over a whole injector sector, it would appear that the smallest dump gap produces a similar level of OGV performance. However, if separate 2-OGV sectors are considered, this is not the case. Between injectors, where there is a flow deficit, λ increases from 0.13 to 0.15, whereas inline with injectors, where there is increased flow, λ falls to 0.1.

It is clear therefore that at the smallest dump gap, the circumferential variation generated by the injectors can modify the flow field as far upstream as the OGV. This is also confirmed by Fig. 11, which shows the circumferential variation of a single OGV exit ($X3$) total pressure measurement located at midannulus height between two OGVs as the combustor is rotated through one injector sector. With a dump gap ratio of 0.8 and a cowl flow of 70%, the measured pressure at OGV exit varies significantly and periodically with the injector location. The magnitude of this variation is relatively large at 7% of the local dynamic pressure. At all other dump gaps and flow rates measured, the pressure is constant to within measurement accuracy. Caution must be exercised here in that this measurement is only at midblade passage, midannulus height, so it cannot be unequivocally concluded that the OGV row does not see the variation in circumferential blockage provided by the combustor at other dump gaps and cowl flows, but it is likely to be the case as the midpassage region is probably the least sensitive to downstream conditions.

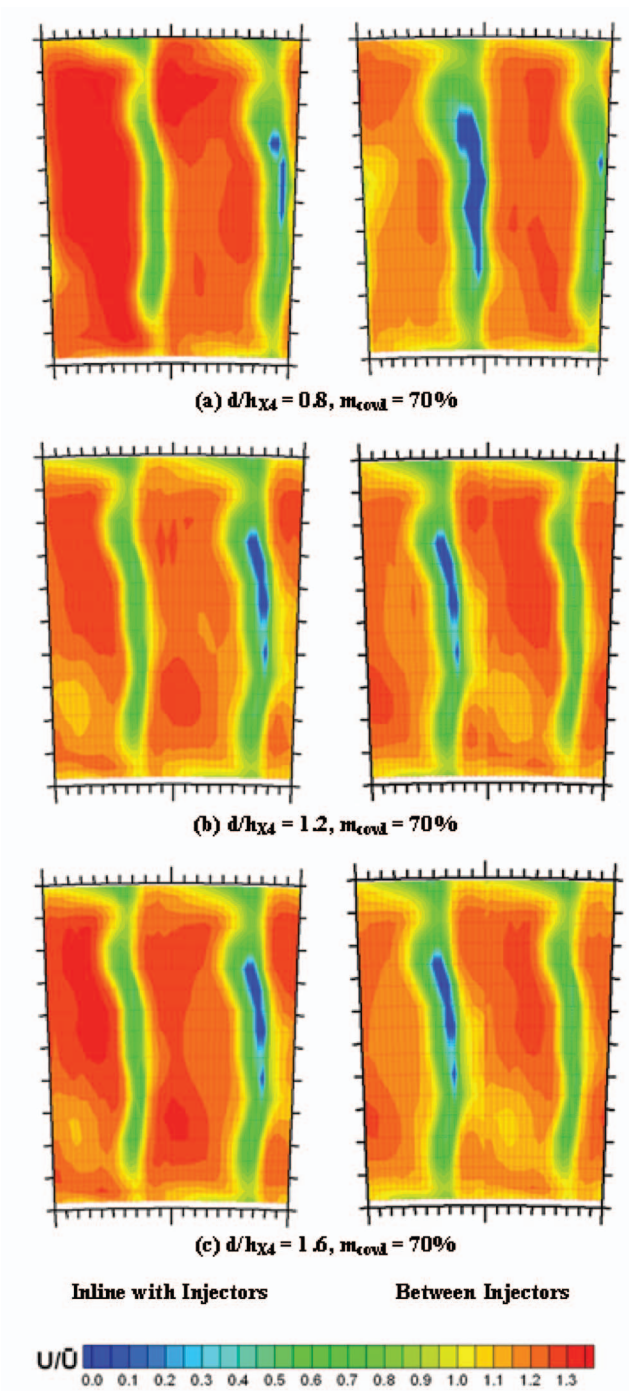

Fig. 10 Axial velocity contours at OGV exit ($X3$) (height =36.6 mm, sector angle $\Delta\theta$=4.5 deg, 2-OGV passages)

Rotor Exit ($X2$)

Figure 12 illustrates clearly that the circumferential variation seen at OGV exit persists into the rotor for a dump gap ratio of 0.8 and a cowl flow of 70%. The plot shows data from a fixed wall static tapping at rotor exit as the combustor was rotated though two sectors. At dump gap ratios of 1.2 and 1.6, there are only slight variations in the measured pressure circumferentially (eight

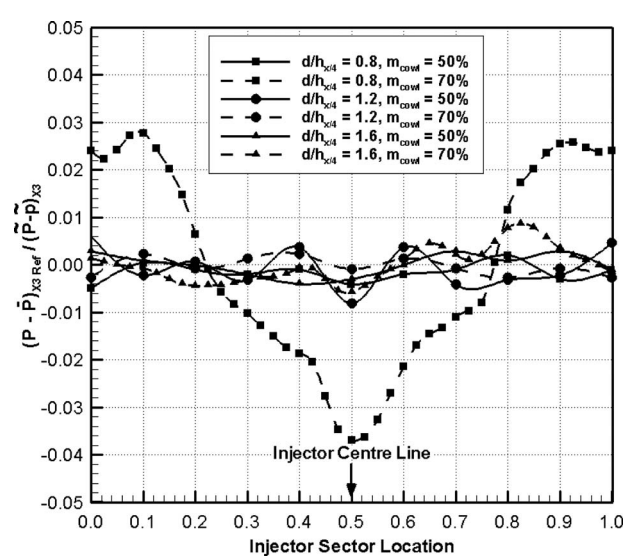

Fig. 11 OGV exit (reference) total pressure

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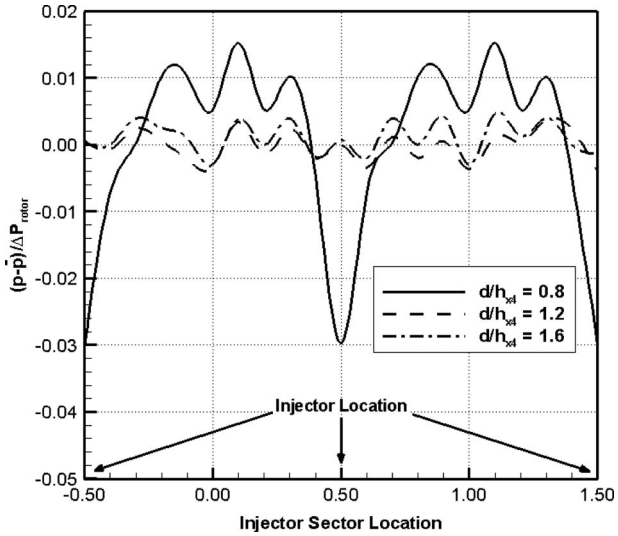

Fig. 12 rotor exit static pressure (m_{cowl}=70%)

peaks and troughs can be seen), with the circumferential resolution in the data being just sufficient to pick up these changes in static pressure, which are associated with the potential field of the OGVs. However, the static pressure for the 0.8 case can be seen to vary significantly from inline to in between injectors, showing a low pressure inline. The variation caused by the cowl flow suction field is much larger than the OGV potential disturbance and is equivalent to approximately 5% of the rotor total pressure rise. This is enough to force the rotor to move locally up or down its characteristic, a fact that is confirmed in the pitch-averaged velocity and total pressure profiles presented in Fig. 13. The forced profiles exhibit a velocity deficit at the tip, a feature not seen in any of the other configurations where the profiles remain unchanged (see also Walker et al. [2]). It is emphasized again that upstream of the rotor, at rig inlet ($X1$) for all configurations, the conditions remained constant.

The fact that the changes in dump gap and downstream flow split can penetrate as far upstream as the rotor and affect its operation is significant, and shows that dump gap design rules to achieve optimum stability and performance for prediffusers will clearly not be transferable between conventional and lean module combustors.

Table 4 Inner annulus performance data (m_{cowl}=70%)

Dump gap (d/h_{X4})	λ_{X2-X5i}	Cp_{X2-X5i}
0.8	0.262	0.71
1.2	0.265	0.72
1.6	0.290	0.70

Annuli ($X5_{i/o}$) and Overall System Performance. Overall system performance was established from measurements in both feed annuli. Traditionally, as the dump gap increases, it would be expected that the turning loss within the dump and hence the annulus loss would reduce. However, at annulus flow splits as low as 15%, the flow entering the annulus is almost entirely comprised of prediffuser boundary layer flow. Already of low quality, this flow is seen to further deteriorate as the dump gap increases and the beneficial blockage effect offered by the flame tube is removed. Indeed, the boundary layers show evidence of separation, which would undoubtedly produce higher levels of turbulence, which would, in turn, increase losses in the shear layer passing around the flame tube head. Tables 4 and 5 detail the mass-weighted performance and confirm that as the dump gap ratio increases from 0.8 to 1.6, the loss increases in both the inner and outer annuli from 26% to 29% of rotor exit dynamic. Further the flow becomes much less uniform as illustrated by normalized axial velocity contours plotted in Figs. 14 and 15. However, associated with the low flow rate, the velocity in the annuli is low (Mach number ~0.02) and consequently the absolute magnitude of the variation is not excessive. For example, Fig. 16 plots contours of total pressure loss coefficient (with respect to rotor exit ($X2$)) at a dump gap ratio of 1.6. This configuration results in the greatest nonuniformity in the velocity field but the figure shows that the total pressure loss varies by less than 2% of its mean value.

Conclusions

The effect on combustion system external aerodynamics of increasing the dump gap from a typical air-spray injector value to

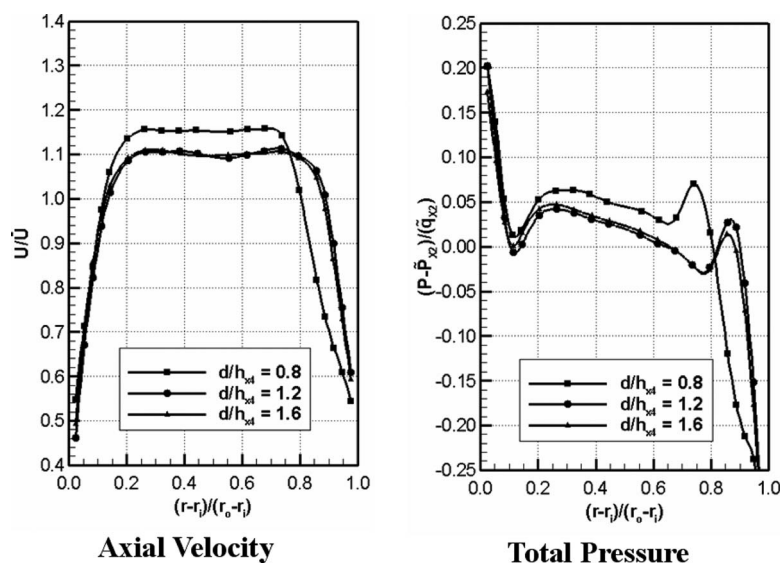

Fig. 13 Pitch averaged profile at rotor exit ($X2$) cowl mass, flow, m_{cowl} =70%

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Table 5 Outer annulus performance data (m_{cowl}=70%)

Dump gap (d/h_{X4})	λ_{X2-X5o}	Cp_{X2-X5o}
0.8	0.263	0.72
1.2	0.268	0.72
1.6	0.291	0.69

values typical of lean module injectors has been studied. Dump gap ratios of 0.8, 1.2, and 1.6, were employed with each test utilizing the same inlet section, IGV, rotor and integrated OGV/prediffuser. The flow fraction of compressor efflux entering the combustor cowl was set to representative lean module levels (50–70%). From the results presented here, the following conclusions can be made.

- At a dump gap ratio (d/h_{X4}) of 0.8, typical of an optimum value for conventional combustor systems, the proximity of the flame tube influences the flow within the prediffuser providing a beneficial blockage effect. However, with an increase to 1.2, possibly needed for access reasons for a lean module injector, the effect is negligible and the prediffuser flow deteriorates. With a further increase to 1.6, the prediffuser no longer feels any influence from the flame tube and shows strong evidence of separation. Hence, at the magnitude of dump gap required for lean module injectors, it is unlikely the prediffuser will "feel" any benefit from the flame tube blockage. This will certainly influence prediffuser design rules, perhaps enforcing more conservative designs.
- The high cowl flow rates of lean module combustors generate a strong circumferential variation in the flame tube head static pressure field. At small dump gaps, this is strong enough to feed upstream through the prediffuser affecting OGV performance and ultimately forcing the rotor. Note that this implies in the present experiment that the compressor moves locally up/down its characteristic as a result of the downstream pressure changes, while keeping the overall flow coefficient constant. However, at larger lean module

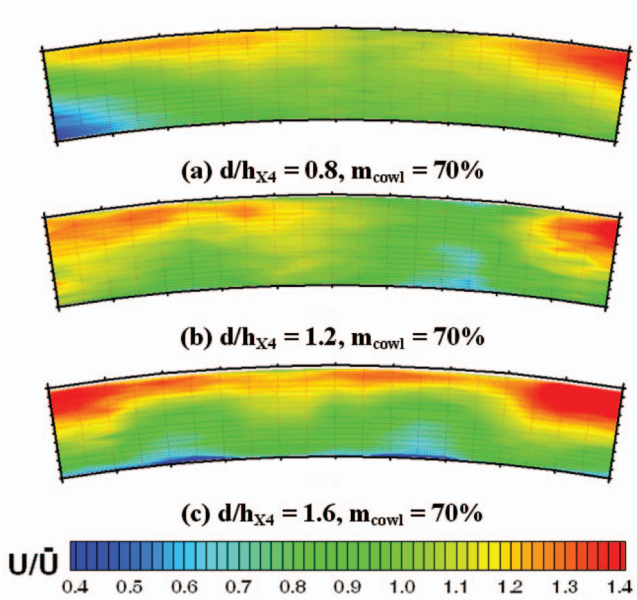

Fig. 15 Axial velocity contours in outer annulus ($X5_o$) (sector angle $\Delta\theta$=18 deg)

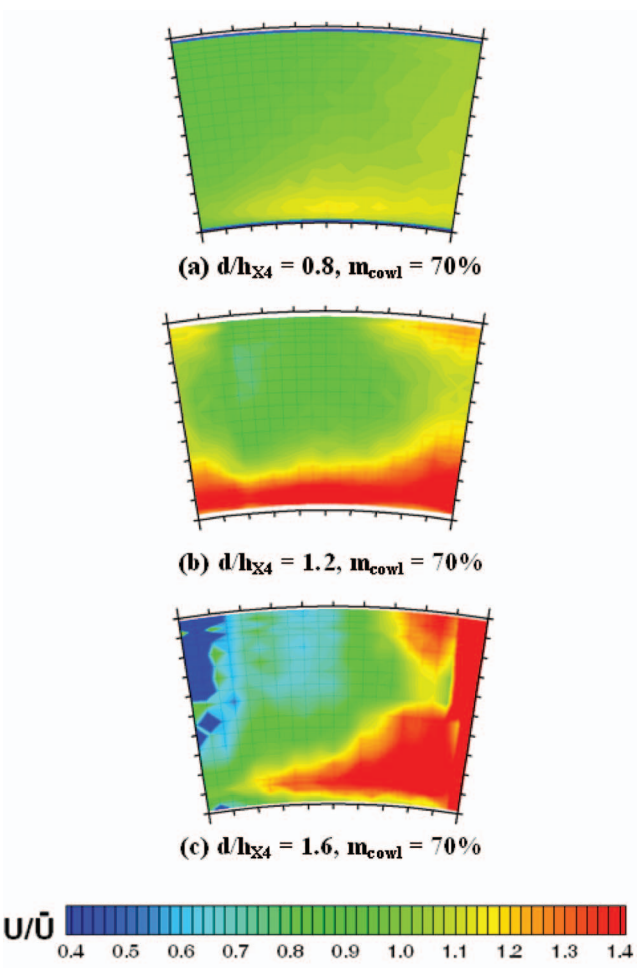

Fig. 14 Axial velocity contours in inner annulus ($X5_i$) (sector angle $\Delta\theta$=18 deg)

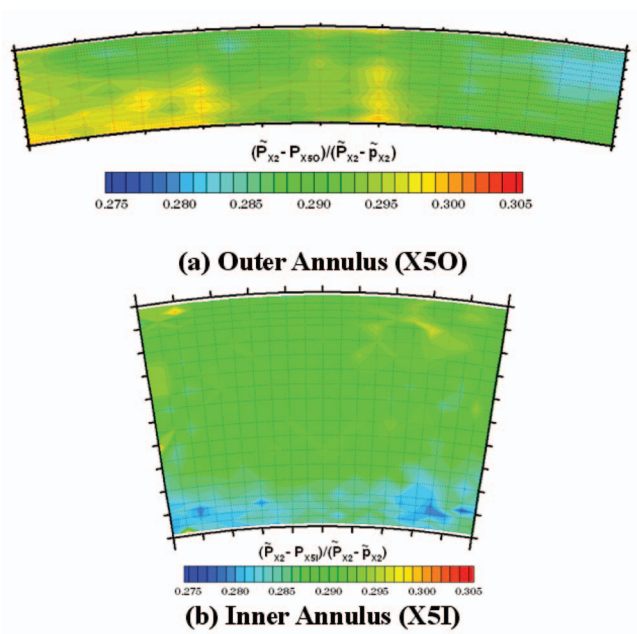

Fig. 16 Loss contours in the feed annuli ($X5$) (d/h_{X4}=1.6, m_{cowl}=70%, sector angle $\Delta\theta$=18 deg)

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dump gaps, the circumferential variation does not penetrate as far upstream as the OGV and the rotor operating condition is unaffected.

- System performance from rotor through to feed annuli is a function of the amount of turning in the dump in conjunction with the condition of the flow at prediffuser exit, particularly the boundary layer. As the dump increases, the turning loss should reduce but here degradation in the prediffuser flow negates this causing a slight increase in loss and a decrease in flow uniformity.
- The present set of experiments has concentrated on a constant flow coefficient case corresponding to the deign condition of the system. It would also be of interest to know the sensitivity of optimum dump gap to off-design operating conditions, and this should be the subject of future work.

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Nomenclature

A = passage area
AR = area ratio
Cp = static pressure rise coefficient
d = distance, dump gap
h = annulus passage height
L = diffuser length
$\dot{m}$ = mass flow rate
N = rotor speed
P, p = local total or static pressure
R = gas constant (perfect gas law)
r = radius relative to rig centerline
r_i, r_o = inner casing, outer casing radius
T, t = total or static temperature
U = rotor midpassage blade speed
V_a = midpassage axial velocity
x = axial distance
W = flame tube depth
α = kinetic energy flux coefficient
ϕ = flow coefficient (va/U)
ρ = density
λ = total pressure loss coefficient

Subscripts

$X1$ = rotor inlet traverse plane
$X2$ = rotor exit/OGV inlet traverse plane
$X3$ = OGV exit traverse plane
$X4$ = prediffuser exit traverse plane
$X5_{i/o}$ = inner/outer annulus traverse plane

Superscripts

$-$ = area weighted spatially averaged mean value
$\sim$ = mass-weighted spatially averaged mean value

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